

Thai Dang Khoa Tran

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PROFESSIONAL SUMMARY

Software engineer and MS Computer Science student (graduating December 2026) with 2+ years of professional full-stack experience building production systems in Java Spring Boot, JavaScript, Vue.js, and PostgreSQL for major Japanese retail chains. Builds data pipelines and serverless AWS applications, ships LLM-integrated features, and writes tests at every level (unit, integration, end-to-end) wired into CI.

EDUCATION

University of the Pacific, Stockton, CA

January 2025 - Present

MS Computer Science | GPA: 3.76 | Expected graduation: December 2026

University of Science, VNU - HCM, Vietnam

2018 - 2022

BS Information Systems

EXPERIENCE

Full-Stack Software Engineer

November 2022 - December 2024

UNICCS Co., Ltd — retail software provider; #1 core-system share among Japan's top 30 drugstore chains

- Architected full-stack features for TAURUS, a core retail ERP (purchase ordering, SKU-level inventory, departmental P&L) used by Don Quijote, Welcia, and Sundrug, with Java Spring Boot backends and JavaScript/Vue.js frontends.
- Reduced query response times by 40% through SQL tuning, indexing, and schema redesign on PostgreSQL and DB2.
- Cut a dashboard stored procedure's runtime from 10 minutes to 30 seconds by isolating the slow join and indexing the key field.
- Deployed and maintained services on AWS, automating build, deployment, and monitoring workflows with Shell scripts.
- Debugged and resolved production issues reported by client teams, tracing defects across the Spring Boot backend and Vue.js frontend.

Tech: Java Spring Boot, JavaScript, Vue.js, PostgreSQL, DB2, SQL, AWS, Shell, HTML, CSS

Software Engineering Intern

December 2021 - June 2022

FPT Telecom — part of FPT Corporation, one of Vietnam's largest tech companies

- Built Node.js/Express REST APIs for real-time monitoring of POP stations across FPT Telecom's Southern Vietnam network.
- Developed React/Next.js dashboards with Highcharts, backed by MongoDB schemas designed for fast monitoring queries.

Tech: React.js, Next.js, Node.js, Express, MongoDB, Highcharts

SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, C++, C#, SQL, HTML, CSS
- **Frameworks:** React.js, Next.js, Vue.js, Node.js, Express, Spring Boot, Tailwind CSS, Hono, D3
- **Databases:** PostgreSQL, MySQL, MongoDB, DB2, SQL Server, Oracle, ClickHouse, Redis
- **Testing & CI/CD:** Vitest, Playwright, unit testing, integration testing, end-to-end testing, GitHub Actions
- **Cloud & Tools:** AWS (Lambda, S3, CloudFront, SNS, IAM), Docker, Vercel, Git, GitHub, Shell, Claude Code
- **Concepts:** OOP, Data Structures & Algorithms, RESTful APIs, Database Design, Data Pipelines/ETL, Serverless, Infrastructure as Code, Agile/Scrum, Machine Learning, NLP, LLM Integration, Prompt Engineering, AI Agents

PROJECTS

US Weather Forecast Pipeline — Personal Project

[Live Demo](#) · [Source Code](#)

- Built an end-to-end data platform (NWS + Open-Meteo → MongoDB → ClickHouse → Redis → Hono API → React) holding 1M+ hourly observations and 1.8M forecast snapshots for 53 US cities, with idempotent incremental loads and data-quality checks.
- Deployed it serverless on AWS as infrastructure as code (SAM/CloudFormation): Lambda + EventBridge scheduled collection and ETL, S3 archive, live dashboard data via CloudFront, SNS alerts, and least-privilege IAM, all at \$0 on the free tier.
- Measured forecast accuracy of NWS and 4 weather models (ECMWF most accurate: 2.1°F average error 1 day ahead), visualized in a D3 dashboard with US map → city → month → day drill-down.
- Directed Claude Code through 4 compare-and-review rounds of 3 parallel builds (Redis caching cut p95 latency 12×); fixed SQL/NoSQL-injection and IAM flaws from security reviews; 107 tests in CI.

Tech: Node.js, Hono, TypeScript, React, D3, MongoDB, ClickHouse, Redis, Docker, AWS (Lambda, S3, CloudFront, SAM), Vitest

CineVibes — Personal Project

[Live Demo](#) · [Source Code](#)

- Built a full-stack movie discussion platform with spoiler-aware comment threads, trending rankings, and CineBot, an in-app AI assistant (Groq-hosted LLM) grounded in the app's MongoDB catalog through a search tool.
- Wrote 57 automated tests (51 Vitest API/integration, 6 Playwright E2E), run with lint and build on every push via GitHub Actions.

Tech: React, Node.js, Express, MongoDB, Vite, JWT Auth, Groq, Vitest, Playwright, GitHub Actions

ML-Assisted Natural Language to SQL Query System — Personal Project

[Live Demo](#) · [Source Code](#)

- Built a CSV-agnostic natural-language-to-SQL system pairing a BERT intent classifier (fine-tuned on 1,250 examples, 14 domains, 5 intents) with a self-built, schema-aware SQL generator and no external LLM APIs.

Tech: Python, BERT (bert-base-uncased), PyTorch, Streamlit, SQLite, Pandas, Matplotlib, Seaborn